



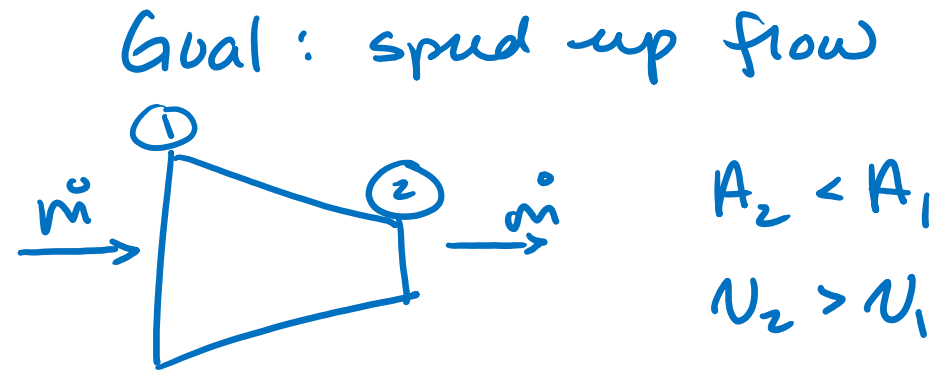
ME 300:
Engineering Thermodynamics

Nozzles and Diffusers

Nozzles

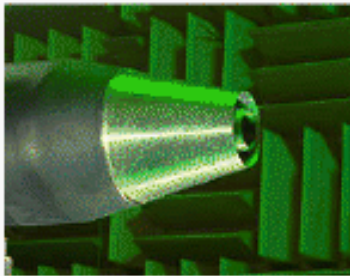


<https://www.worldsbestbrassnozzle.com/products/water-jet-nozzle-solid-brass-sweeper-sprayer>

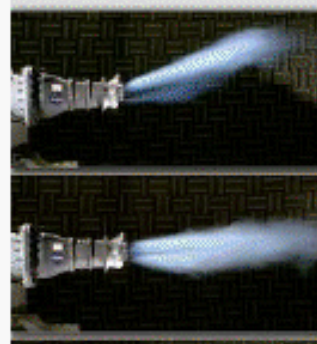


→ Passive: $\dot{W} = 0$

Adiabatic: $\dot{Q} = 0$



Co-annular

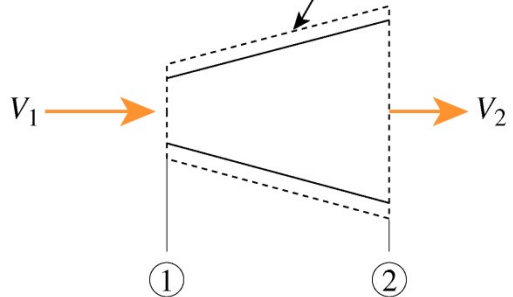
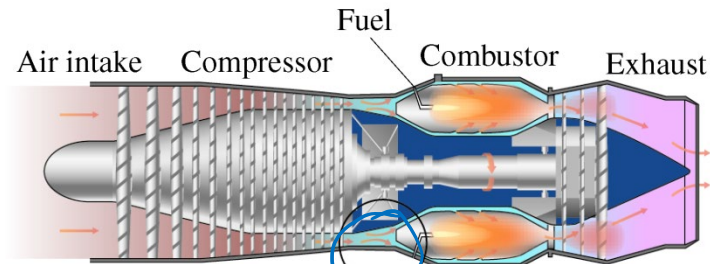


Maneuvering

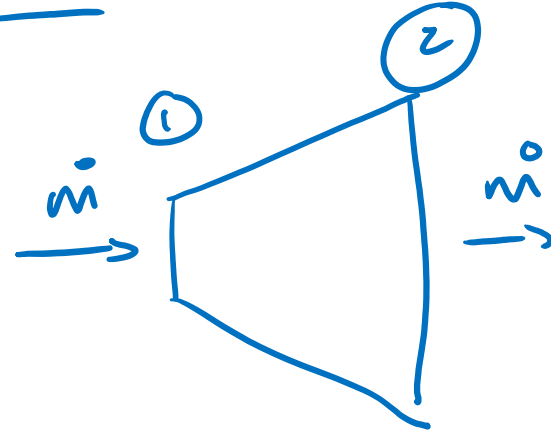


Convergent

Diffusers



Goal: slow flow down



$$\left. \begin{array}{l} A_2 > A_1 \\ v_2 < v_1 \end{array} \right\}$$

→ passive: $\dot{W} = 0$

→ adiabatic: $\dot{Q} = 0$

Steady flow energy equation

$$\dot{m} (\Delta h + \Delta ke + \Delta pe) = \cancel{\dot{Q}} - \cancel{\dot{W}}$$

$$\Rightarrow \Delta h + \Delta ke = 0$$

$$(h_2 - h_1) + \frac{1}{2}(v_2^2 - v_1^2) = 0$$